

Principal Geodesic Analysis on the Sphere

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Abstract

Principal component analysis (PCA) identifies dominant linear modes of variation in Euclidean data by diagonalizing a covariance operator. On curved spaces, such as the round sphere S^2 , this linear structure is no longer available. Following Fletcher, Lu, Pizer, and Joshi, principal geodesic analysis (PGA) replaces linear subspaces by geodesic submanifolds obtained from the exponential and logarithm maps at an intrinsic mean. In this note we work out the geometry of S^2 , review Euclidean PCA, describe PGA in the Riemannian setting, and then analyze in detail the special case of data concentrated near a great circle. We complement the theory with a simple numerical experiment based on Gaussian samples in a tangent space, illustrating how PGA retains a stable intrinsic description of variation while Euclidean PCA becomes increasingly influenced by the ambient embedding as the spread of the data grows.

1 Introduction

Principal component analysis (PCA) is a classical method for finding dominant directions of variation in Euclidean data. Given points in $\mathbb{R}^n$, PCA diagonalizes the sample covariance matrix and produces orthonormal directions along which the variance is maximal. In differential geometry, dominant directions arise in a similar way by diagonalizing symmetric bilinear forms such as the second fundamental form with respect to the first fundamental form. For example, diagonalizing the shape operator produces principal curvatures and principal directions on a surface. This parallel suggests that the linear structure of PCA should have a natural analogue on curved spaces.

In many applications data do not lie in a vector space but on a Riemannian manifold. One basic example that we wish to explore is the unit sphere $S^2 \subset \mathbb{R}^3$ with its standard metric. For such data sets, linear PCA in the ambient $\mathbb{R}^3$ ignores curvature and can give misleading results. Principal geodesic analysis (PGA) is a generalization of PCA which respects the Riemannian structure. We build off of the work of Fletcher, Lu, Pizer, and Joshi [2] in exploring this case.

2 Geometry of the Sphere

Let

$$S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$$

be the unit sphere with the metric induced from the Euclidean inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^3$.

The sphere can be realized as the level set of the function

$$F(x) = \|x\|^2 - 1 = x_1^2 + x_2^2 + x_3^2 - 1.$$

Then $\nabla F(x) = 2x$, so at a point $p \in S^2$ the gradient $\nabla F(p)$ is a normal vector. The tangent space of the level set is always orthogonal to the gradient, so

$$T_p S^2 = \{v \in \mathbb{R}^3 : \langle v, \nabla F(p) \rangle = 0\} = \{v \in \mathbb{R}^3 : \langle v, p \rangle = 0\}.$$

The Riemannian distance $d(p, q)$ between $p, q \in S^2$ is the length of the shortest geodesic joining them. On the unit sphere, this shortest geodesic is just the shorter arc of the great circle lying in the plane spanned by p and q . If we denote by θ the angle between p and q in $\mathbb{R}^3$, then $\cos(\theta) = \langle p, q \rangle$ and the length of this arc is θ . Therefore

$$d(p, q) = \arccos(\langle p, q \rangle) \in [0, \pi].$$

Geodesics on S^2 are the great circles, that is, intersections of S^2 with two-dimensional linear subspaces of $\mathbb{R}^3$. Fix $p \in S^2$ and a tangent vector $v \in T_p S^2$ with $\|v\| = 1$. Consider

$$\gamma(t) = \cos(t)p + \sin(t)v.$$

First, $\gamma(t)$ remains on S^2 :

$$\|\gamma(t)\|^2 = \langle \cos t p + \sin t v, \cos t p + \sin t v \rangle = \cos^2 t + \sin^2 t = 1.$$

Second, the velocity is always tangent:

$$\gamma'(t) = -\sin t p + \cos t v,$$

and

$$\langle \gamma(t), \gamma'(t) \rangle = \langle \cos t p + \sin t v, -\sin t p + \cos t v \rangle = -\cos t \sin t + \sin t \cos t = 0.$$

Thus $\gamma'(t) \in T_{\gamma(t)} S^2$. Finally, the acceleration

$$\gamma''(t) = -\cos t p - \sin t v = -\gamma(t)$$

is always normal to the sphere (it is a multiple of the radius vector). Hence

$$\|\gamma'(t)\|^2 = \sin^2 t + \cos^2 t = 1,$$

so γ is a unit-speed geodesic with $\gamma(0) = p$ and $\gamma'(0) = v$.

If v is an arbitrary nonzero tangent vector, write $v = \|v\|\hat{v}$ with $\|\hat{v}\| = 1$, and set

$$\gamma(t) = \cos(t\|v\|)p + \sin(t\|v\|)\hat{v}.$$

Then the same computation shows that γ is a geodesic with $\gamma(0) = p$ and $\gamma'(0) = v$, and γ has constant speed $\|v\|$.

By definition, the exponential map at p sends a tangent vector v to $\gamma(1)$, where γ is the geodesic with initial data (p, v) . Using the formula above we obtain

$$\text{Exp}_p : T_p S^2 \rightarrow S^2, \quad \text{Exp}_p(v) = \cos(\|v\|)p + \sin(\|v\|)\frac{v}{\|v\|}$$

for $v \neq 0$ and $\text{Exp}_p(0) = p$.

Conversely, fix $p \in S^2$ and consider a point $x \in S^2$ with $x \neq -p$. There is a unique shortest geodesic from p to x , and it has length

$$\theta = d(p, x) = \arccos(\langle p, x \rangle) \in (0, \pi).$$

We now solve for the tangent vector v such that $x = \text{Exp}_p(v)$ and $\|v\| = \theta$.

By the exponential map formula,

$$x = \text{Exp}_p(v) = \cos(\theta) p + \sin(\theta) \frac{v}{\theta}$$

since $\|v\| = \theta$. Rearranging gives

$$x - \cos(\theta) p = \sin(\theta) \frac{v}{\theta}.$$

Taking norms on both sides and using $\|x\| = \|p\| = 1$ we find

$$\|x - \cos(\theta) p\|^2 = 1 + \cos^2 \theta - 2 \cos \theta \langle x, p \rangle = 1 + \cos^2 \theta - 2 \cos^2 \theta = 1 - \cos^2 \theta = \sin^2 \theta,$$

so $\|x - \cos(\theta) p\| = \sin \theta$. Thus

$$\frac{v}{\theta} = \frac{x - \cos(\theta) p}{\sin(\theta)} \implies v = \theta \frac{x - \cos(\theta) p}{\sin(\theta)}.$$

We define

$$\text{Log}_p(x) = \theta \frac{x - \cos(\theta) p}{\sin(\theta)}, \quad \theta = \arccos(\langle p, x \rangle).$$

The map Log_p is defined on $S^2 \setminus \{-p\}$ and is the inverse of Exp_p restricted to the open geodesic ball of radius π around p .

3 Principal Component Analysis

Let $x_1, \dots, x_m \in \mathbb{R}^n$ be data points. Assume they are centered,

$$\frac{1}{m} \sum_{i=1}^m x_i = 0.$$

The sample covariance matrix is

$$C = \frac{1}{m} \sum_{i=1}^m x_i x_i^\top.$$

This is a real symmetric $n \times n$ matrix, so there exists an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors $v_1, \dots, v_n$ with eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0.$$

For a unit vector $v \in \mathbb{R}^n$, the variance of the data in direction v is

$$Q(v) = v^\top C v = \frac{1}{m} \sum_{i=1}^m \langle x_i, v \rangle^2.$$

To find the first principal component, one maximizes Q over the unit sphere $\{v : \|v\| = 1\}$. Introducing a Lagrange multiplier and differentiating shows that critical points of Q under this constraint are exactly the unit eigenvectors of C . Among these, eigenvectors corresponding to the largest eigenvalue λ_1 maximize Q .

The subspace $V_r = \text{span}\{v_1, \dots, v_r\}$ spanned by the first r eigenvectors gives the best r -dimensional linear approximation of the data in the least squares sense. More precisely, for any r -dimensional subspace $W \subset \mathbb{R}^n$, let Π_W denote the orthogonal projection onto W and define

$$E(W) = \sum_{i=1}^m \|x_i - \Pi_W x_i\|^2.$$

A standard form of the Eckart–Young theorem (see, for example, any text on PCA) states that

$$E(W) \geq m \sum_{k=r+1}^n \lambda_k,$$

with equality if and only if $W = V_r$. In other words, V_r is the unique subspace that minimizes the total squared reconstruction error. We will not prove this general result here, since our focus is on how the PCA structure is seen on the sphere, but we will use the fact that Euclidean PCA can be characterized in this way.

4 Intrinsic Means and PGA on S^2

We now adapt the PCA construction to the Riemannian setting, following the framework of Fletcher, Lu, Pizer, and Joshi [2], and then specialize everything to the sphere S^2 using the explicit geometry described above.

Let (M, g) be a Riemannian manifold and let $x_1, \dots, x_N \in M$ be given data points. In [2], the intrinsic (Fréchet) mean of the x_i is defined as a point $\mu \in M$ minimizing the sum of squared distances

$$\mu = \operatorname{argmin}_{x \in M} \sum_{i=1}^N d(x, x_i)^2.$$

This generalizes the Euclidean notion of the mean as the minimizer of the sum of squared Euclidean distances. In practice, Fletcher et al. compute such a mean numerically by iteratively moving along geodesics in directions suggested by the data, using the exponential and logarithm maps.

Once a mean μ has been found, the data are mapped to the tangent space $T_\mu M$ using the Riemannian logarithm map:

$$u_i = \operatorname{Log}_\mu(x_i) \in T_\mu M.$$

In an orthonormal basis of $T_\mu M$, one defines the covariance operator

$$S = \frac{1}{N} \sum_{i=1}^N u_i u_i^\top,$$

just as in the Euclidean PCA setting. This operator is symmetric and can be diagonalized. Let $v_1, \dots, v_d$ be an orthonormal eigenbasis of $T_\mu M$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d \geq 0$, where

$d = \dim M$. Fletcher et al. interpret these v_k as principal directions in the tangent space and then map them back to the manifold using the exponential map.

More precisely, the first principal geodesic is defined in [2] as the image of the one-dimensional subspace $\text{span}\{v_1\}$ under the exponential map at μ :

$$\Gamma_1 = \{\text{Exp}_\mu(tv_1) : t \in \mathbb{R}\}.$$

Higher-dimensional principal geodesic submanifolds are obtained similarly from $\text{span}\{v_1, \dots, v_k\}$. This is the manifold analogue of the Euclidean principal subspace $V_r = \text{span}\{v_1, \dots, v_r\}$: in both cases one diagonalizes a covariance operator and keeps the directions corresponding to the largest eigenvalues, but in the manifold case those directions are realized as geodesics rather than straight lines.

In our case $M = S^2$ with the round metric studied in the previous section. The tangent spaces $T_\mu S^2$ are two-dimensional, so there are only two eigenvalues. The image of $\text{span}\{v_1\}$ under Exp_μ is a great circle through μ , and the image of $\text{span}\{v_1, v_2\}$ is all of S^2 . Thus for S^2 the only nontrivial PGA object of study is the first principal geodesic, which is a great circle chosen to capture the dominant variation of the data with respect to the spherical metric.

5 Data Along a Great Circle

We now consider in detail the special case where the data lie along a single great circle on S^2 . This example allows us to compute both Euclidean PCA and PGA explicitly and to see clearly what information PGA adds. The setup is analogous in spirit to examples in [2], but here we use the explicit formulas for S^2 from the previous section.

Fix a unit vector $p \in S^2$ and a unit tangent vector $w \in T_p S^2$, so that $\langle p, w \rangle = 0$ and $\|w\| = 1$. The great circle determined by (p, w) is the geodesic

$$\gamma(t) = \cos(t)p + \sin(t)w, \quad t \in \mathbb{R}.$$

We choose data points of the form

$$x_i = \gamma(t_i) = \cos(t_i)p + \sin(t_i)w, \quad i = 1, \dots, N,$$

for some parameters t_i . We assume that the parameters are symmetric around 0, in the sense that whenever t appears among the t_i , so does $-t$ with the same multiplicity.

We first carry out Euclidean PCA on the data x_i regarded as vectors in $\mathbb{R}^3$. The Euclidean mean is

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i = \frac{1}{N} \sum_{i=1}^N (\cos(t_i)p + \sin(t_i)w) = \alpha p + \beta w,$$

where

$$\alpha = \frac{1}{N} \sum_{i=1}^N \cos(t_i), \quad \beta = \frac{1}{N} \sum_{i=1}^N \sin(t_i).$$

By symmetry of the t_i , the sum of the sines is zero. Thus $\beta = 0$, and

$$\bar{x} = \alpha p, \quad \alpha = \frac{1}{N} \sum_{i=1}^N \cos(t_i).$$

If not all t_i are zero, then at least one $|t_i| > 0$ so $\cos(t_i) < 1$ for that index, and hence $0 < \alpha < 1$. Therefore $\bar{x}$ lies strictly inside the unit ball and not on S^2 . A common practical fix could be to project $\bar{x}$ back to S^2 by normalization, obtaining $\bar{x}/\|\bar{x}\| = p$, so the projected mean coincides with p , but the Euclidean optimization problem and the spherical one are different.

The covariance structure in $\mathbb{R}^3$ is also straightforward. Each x_i lies in the two-dimensional plane spanned by p and w , so all variation is contained in that plane and there is no variation in the direction $p \times w$. Writing

$$x_i - \bar{x} = (\cos(t_i) - \alpha)p + \sin(t_i)w,$$

we see that the covariance matrix has rank at most 2 and that its eigenvectors lie in the span of $\{p, w\}$. If the t_i are small and symmetric, we can expect that the principal direction in $\mathbb{R}^3$ will be close to w .

Intrinsic mean and PGA on S^2

We now analyze the same data from the viewpoint of PGA on S^2 , using the explicit exponential and logarithm maps. The first step, following [2], is to find an intrinsic mean, that is, a minimizer of

$$\mu = \operatorname{argmin}_{x \in S^2} \sum_{i=1}^N d(x, x_i)^2.$$

In our special configuration, the data lie on the great circle γ , and the parameters t_i are symmetric around 0. Intuitively, the point $p = \gamma(0)$ is the “center” of the configuration along the circle, so it is natural to expect that $\mu = p$.

We can make this more precise by using the explicit form of the logarithm map. For each i we have

$$x_i = \gamma(t_i) = \operatorname{Exp}_p(t_i w),$$

since γ is the geodesic with initial data (p, w) . Therefore

$$\operatorname{Log}_p(x_i) = t_i w.$$

The characterization of intrinsic means in terms of the logarithm map used in [2] says that at a critical point μ one should have

$$\sum_{i=1}^N \operatorname{Log}_\mu(x_i) = 0.$$

At $\mu = p$ this sum is

$$\sum_{i=1}^N \operatorname{Log}_p(x_i) = \sum_{i=1}^N t_i w = \left(\sum_{i=1}^N t_i \right) w,$$

which vanishes by symmetry of the t_i .

Having identified the mean, we map the data to the tangent space $T_p S^2$:

$$u_i = \operatorname{Log}_p(x_i) = t_i w.$$

Thus u_i lie on the one-dimensional subspace $\text{span}\{w\} \subset T_p S^2$. The covariance operator on $T_p S^2$ is

$$S = \frac{1}{N} \sum_{i=1}^N u_i u_i^\top = \frac{1}{N} \sum_{i=1}^N (t_i w)(t_i w)^\top = \left(\frac{1}{N} \sum_{i=1}^N t_i^2 \right) w w^\top.$$

This operator has one nonzero eigenvalue

$$\lambda = \frac{1}{N} \sum_{i=1}^N t_i^2$$

with eigenvector w , and one zero eigenvalue with eigenvectors orthogonal to w . In particular, the first principal direction in the tangent space is exactly w .

The associated principal geodesic is the image of the line $\text{span}\{w\}$ under Exp_p :

$$\Gamma_1 = \{\text{Exp}_p(tw) : t \in \mathbb{R}\} = \{\cos(t)p + \sin(t)w : t \in \mathbb{R}\} = \gamma(\mathbb{R}).$$

Thus in this example, PGA exactly recovers the true one-dimensional manifold structure of the data, identifying both the intrinsic mean p and the great circle along which the data lie. This is the simplest case where the PGA construction on S^2 reduces to exponentiating a one-dimensional principal subspace in a tangent plane.

6 Numerical Experiment and Comparison with Euclidean PCA

To see how this picture changes when the data no longer lie on a single geodesic but occupy a neighborhood in S^2 , we implemented a simple numerical experiment following the PGA framework of [2]. The code is contained in a Colab notebook [3].

We fix $\mu = (0, 0, 1) \in S^2$ and choose an orthonormal basis $\{e_1, e_2\}$ of $T_\mu S^2$, so that any tangent vector can be written as $u = ae_1 + be_2$. For each parameter $s > 0$ we define data in the tangent plane by sampling

$$(a, b) \sim \mathcal{N}((0, 0), \Sigma_s), \quad \Sigma_s = \begin{pmatrix} (s\sigma_1)^2 & 0 \\ 0 & (s\sigma_2)^2 \end{pmatrix},$$

with fixed base scales $\sigma_1 = 0.3$ and $\sigma_2 = 0.1$. Thus the relative spread in the e_1 -direction is three times larger in standard deviation than in the e_2 -direction, so most of the variation in the tangent plane is along e_1 , while s controls how far the samples wander from the origin along geodesics.

For each value of s we then:

- perform PGA at μ by mapping x_i back to $T_\mu S^2$ using $u_i = \text{Log}_\mu(x_i)$ and diagonalizing the covariance operator on $T_\mu S^2$;
- perform standard Euclidean PCA on the embedded data $x_i \in \mathbb{R}^3$ by diagonalizing the sample covariance in $\mathbb{R}^3$.

In both cases we record the fraction of variance explained by the first principal component. For Euclidean PCA we also record the variance in the third principal direction, which corresponds to motion orthogonal to the sphere. Finally, as a reference configuration without a preferred direction, we repeat the computations for $N = 2000$ points sampled uniformly at random on S^2 .

The output of this experiment is shown in Table 1.

Table 1: Representative simulation results for PGA and Euclidean PCA on S^2 . The columns show the fraction of variance explained by the first principal component for PGA and PCA, and the fraction of variance explained by the third PCA component.

case	spread s	PGA var ₁	PCA var ₁	PCA var ₃
Gaussian	0.10	0.897	0.889	0.004
Gaussian	0.30	0.896	0.880	0.010
Gaussian	0.60	0.896	0.858	0.025
Gaussian	1.00	0.894	0.822	0.051
Gaussian	1.40	0.893	0.781	0.083
Gaussian	1.80	0.892	0.732	0.123
Uniform	—	0.504	0.337	0.330

7 Discussion

Principal geodesic analysis offers a natural extension of PCA to Riemannian manifolds. On the sphere S^2 the construction is especially transparent: the exponential and logarithm maps are given by elementary trigonometric formulas, intrinsic means are defined using spherical distance, and principal directions in the tangent plane are lifted back to the sphere along great circles. The example of data concentrated near a geodesic, together with the numerical experiment based on Gaussian samples in a tangent space, shows that PGA retains a stable description of the main direction of variation, while Euclidean PCA becomes increasingly influenced by the embedding as the data spread over a curved region.

Beyond the round sphere, the same ideas apply to more complicated spaces where data naturally live on nonlinear manifolds as in [2]. In those settings one no longer has explicit formulas for the exponential and logarithm maps, and both the computation of intrinsic means and the evaluation of principal geodesics become numerical optimization problems based on the Fréchet energy and its Riemannian gradient. For a detailed account of these algorithms in more general manifolds, we refer back to Fletcher, Lu, Pizer, and Joshi [2].

References

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